

# PAPER\_1390\_TROUTON\_NOBLE

*Daniel T. Murphy*

## PAPER\_1390 — UQFF Resolution of the Trouton-Noble Torque Paradox (CLOSED — Dispatch Whitepaper)

**Author:** Daniel T. Murphy

**Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

**Tier:** B — Foundational / Special Relativity (CLOSED)

**Date:** June 16, 2026

**Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA)

**Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "trouton_noble"})`

**Calculator surface:** `calculate_paradox({"paradox": "trouton_noble"})`

**Closure helper:** `_l96_uqff_axiom_trouton_noble_closure()`

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### The Paradox

A charged parallel-plate capacitor moving through the ether (or any inertial frame) classically experiences a torque tending to rotate it perpendicular to its motion. Experimentally, no torque is observed.

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### UQFF Closed Identity

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Net torque on moving capacitor = 0 EXACT via  $F_U = 1$  global ledger symmetry

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$F_U = 1$  global ledger symmetry: any net torque would break the global normalization  $F_U = 1$ , which is forbidden by PAPER\_646. Therefore  $\tau_{\text{net}} = 0$  EXACT.

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### Physical Interpretation

The classical 'Trouton-Noble torque' computation double-counts the EM stress tensor by ignoring the  $F_U = 1$  global constraint. Once  $F_U = 1$  is enforced, the torque cancels EXACTLY.

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### Live Calculator Output

```
``python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "trouton_noble"})["value"]
``
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_trouton_noble_closure` in `uqff_pure_calculator.py` for the full field list.

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## NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER\_646, PAPER\_597, PAPER\_1051, etc.). **Zero free parameters.**

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## Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_trouton_noble_closure`
- Dispatch keys: `trouton_noble`, `trouton_noble_paradox`
- Related foundational papers: PAPER\_646 ( $F_U B_i / F_U = 1$  ledger), PAPER\_597 (negative-time dual existence), PAPER\_1183 (paradox consolidation).

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